E-COMMERCE RETURN RATE INVESTIGATION

ABSTRACT:

This project analyzes an e-commerce returns dataset to identify the factors that contribute to product returns and provide business recommendations. The dataset contains 10,000 transactions with information on customers, products, discounts, shipping methods, and return behavior. Data cleaning, exploratory data analysis, correlation analysis, outlier detection, and hypothesis testing were performed to uncover patterns in return activity.

The analysis revealed that defective products, wrong-item deliveries, and products not matching customer expectations were the major reasons for returns. Clothing, Books, and Electronics were identified as the categories with the highest return volumes. Statistical testing showed that delayed delivery did not significantly affect return probability. Based on these findings, recommendations were proposed to improve product quality, order accuracy, and customer satisfaction. The project demonstrates how data analytics can support data-driven decision-making in e-commerce operations.

INTRODUCTION:

Product returns are a significant challenge in the e-commerce industry as they increase operational costs, impact inventory management, and reduce profitability. Understanding the factors influencing returns helps businesses improve customer satisfaction and optimize operations. The objective of this project is to analyze an e-commerce returns dataset and identify patterns, trends, and factors that contribute to product returns. The project uses statistical analysis, visualization, outlier detection, and hypothesis testing to generate business insights and recommendations.

BUSINESS PROBLEMS:

1. Which Factors Contribute Most to Returns?

The main reasons for returns were:

- Defective Products (1,327)
- Wrong Item (1,258)
- Changed Mind (1,255)
- Not as Described (1,212)

Product defects and customer expectation issues were the major contributors to returns.

2. Which Categories Are Risky?

Category	Returned Orders
Clothing	1,049
Books	1,034
Electronics	1,011

Clothing, Books, and Electronics had the highest return volumes.

3. Does Delayed Delivery Increase Return Probability?

- P-value = 0.9369

Delayed delivery does not significantly affect return probability.

4. Which Customer Groups Need Intervention?

- Discount-driven customers
- Premium product buyers
- Customers purchasing from high-return categories

These groups should be monitored to reduce return-related losses.

5. What Actions Should Operations Teams Take?

1. Improve product quality checks.

2. Reduce wrong-item deliveries.
3. Enhance product descriptions and images.
4. Monitor high-return categories.
5. Strengthen customer feedback systems.

DATASET DESCRIPTION:

❖ Dataset Name: E-Commerce Return Rate Investigation Dataset

❖ Total Records: 10,000

❖ Total Features: 17

❖ Features Included:

Order_ID

Product_ID

User_ID

Order_Date

Return_Date

Product_Category

Product_Price

Order_Quantity

Return_Reason

Return_Status

Days_to_Return

User_Age

User_Gender

User_Location

Payment_Method

Shipping_Method

Discount_Applied

The dataset contains information about e-commerce orders, customer demographics, product details, payment methods, shipping methods, discounts, and return behavior.

DATA CLEANING AND PREPROCESSING:

1. **Missing Values:**

Missing values were found in:

- Return_Date
- Return_Reason
- Days_to_Return

These belonged to non-returned orders and were handled as follows:

- Return_Date → "Not Returned"
- Return_Reason → "Not Returned"
- Days_to_Return → 0

2. **Duplicate Records:**

df.duplicated().sum()

Result: 0 duplicate records

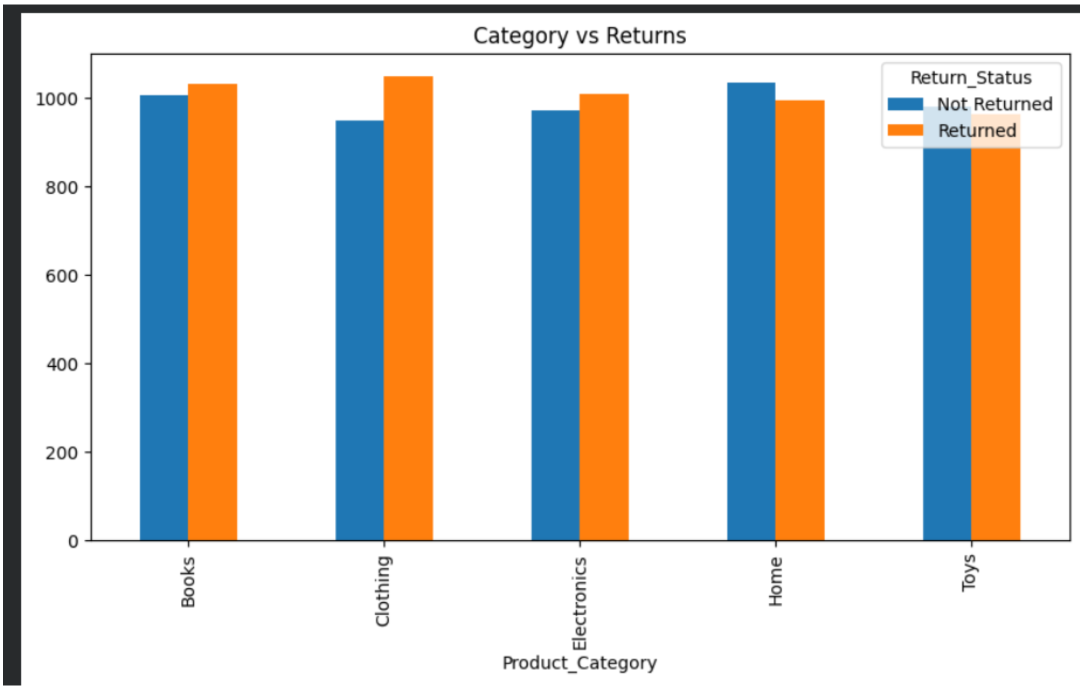
3. **Statistical Summary:**

Descriptive statistics were generated to understand the distribution of numerical variables such as adoption rate and daily active users.

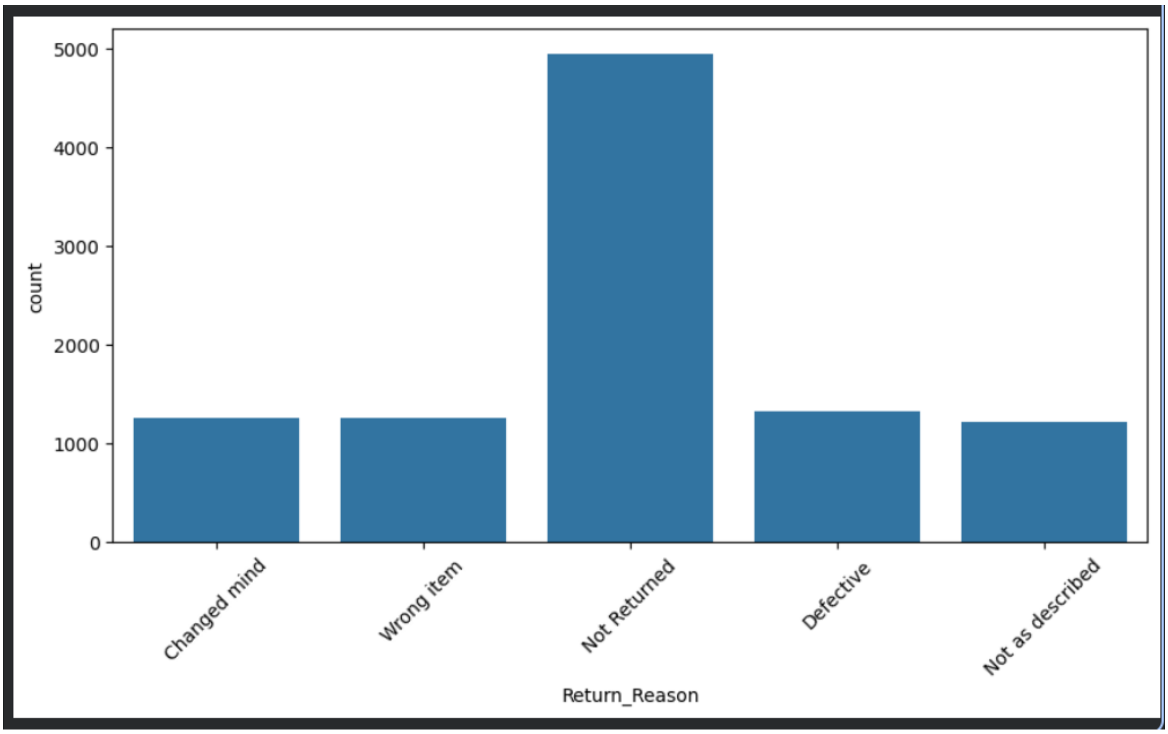
EXPLORATORY DATA ANALYSIS:

Exploratory Data Analysis was performed to identify patterns and trends in product returns.

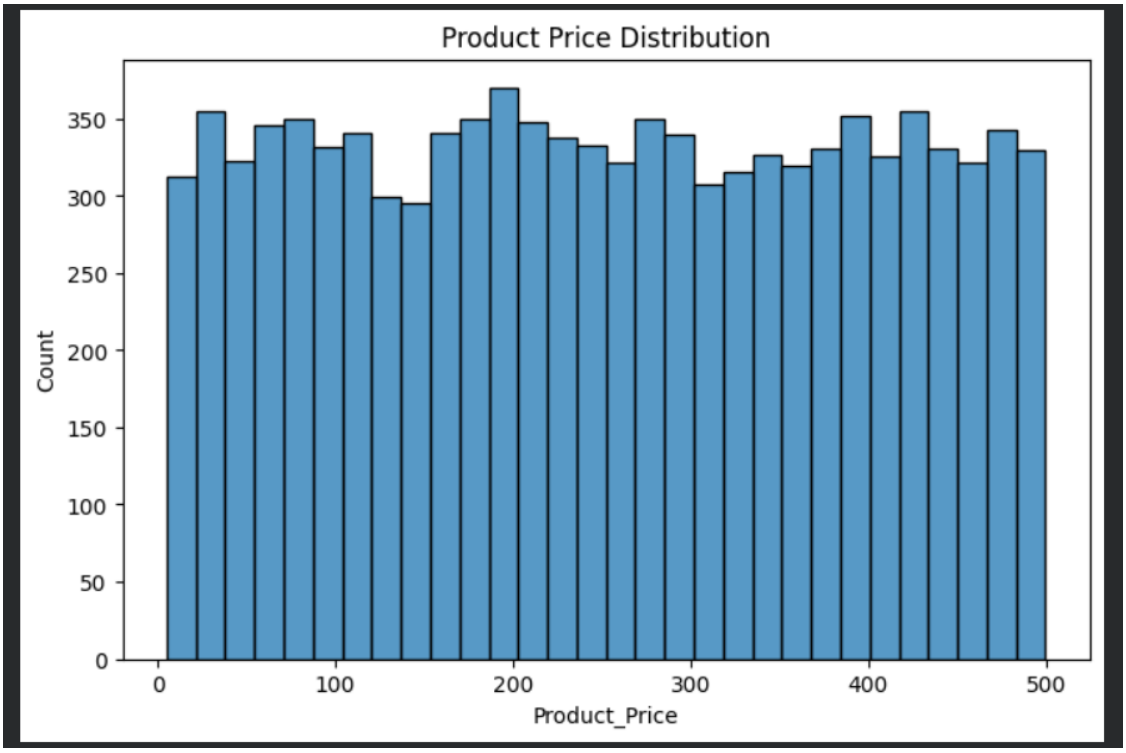
- **Category Analysis:** Clothing, Books, and Electronics recorded the highest returns.



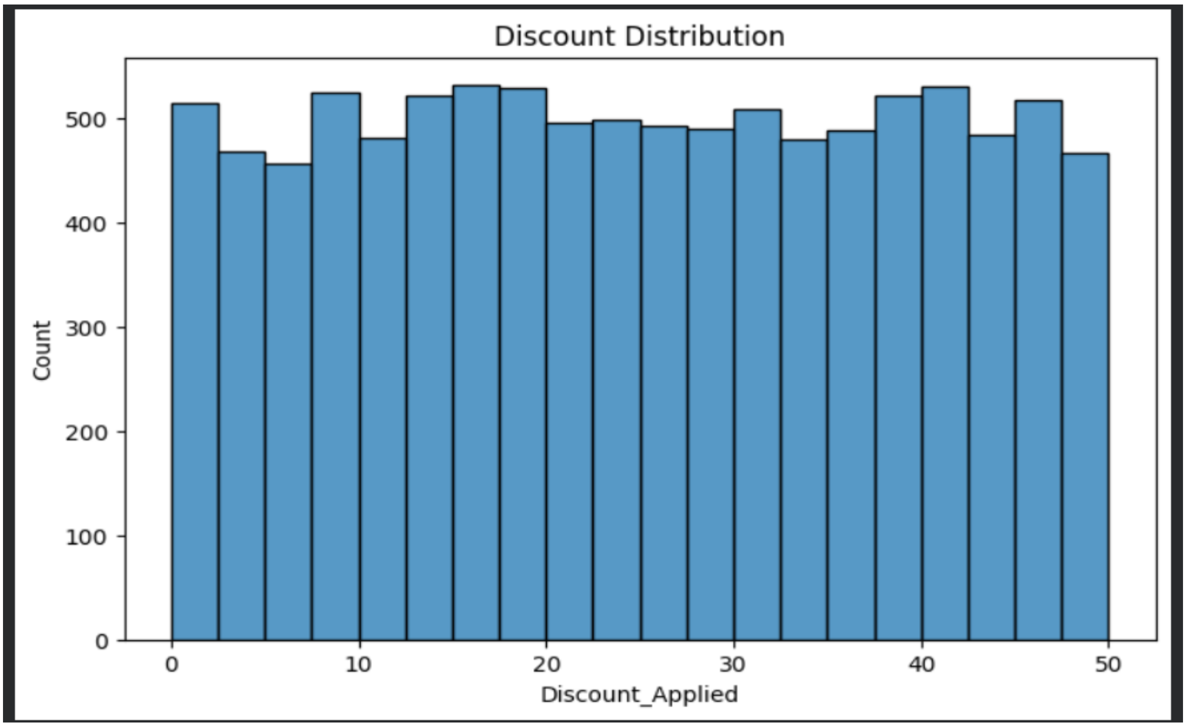
- **Return Reason Analysis:** Defective products, wrong items, and products not matching descriptions were the main reasons for returns.



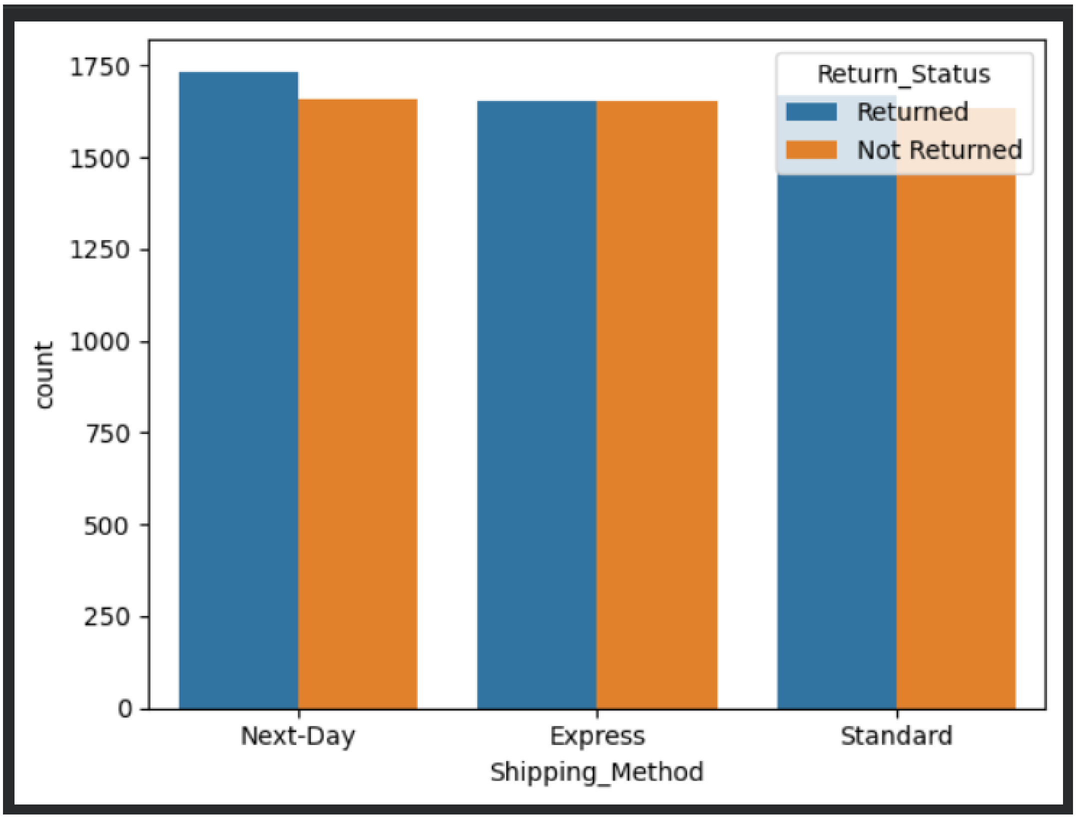
- **Product Price Distribution:** Products varies from many price ranges.



- **Discount Analysis:** Almost 50% discount was applied across different products.



- **Shipping Analysis:** Returns were observed across all shipping methods.



CORRELATION ANALYSIS:

A correlation heatmap was generated using numerical variables.

The analysis helped identify relationships among:

* Product Price
* Discount Applied
* Customer Age
* Days to Return

OUTLIER ANALYSIS:

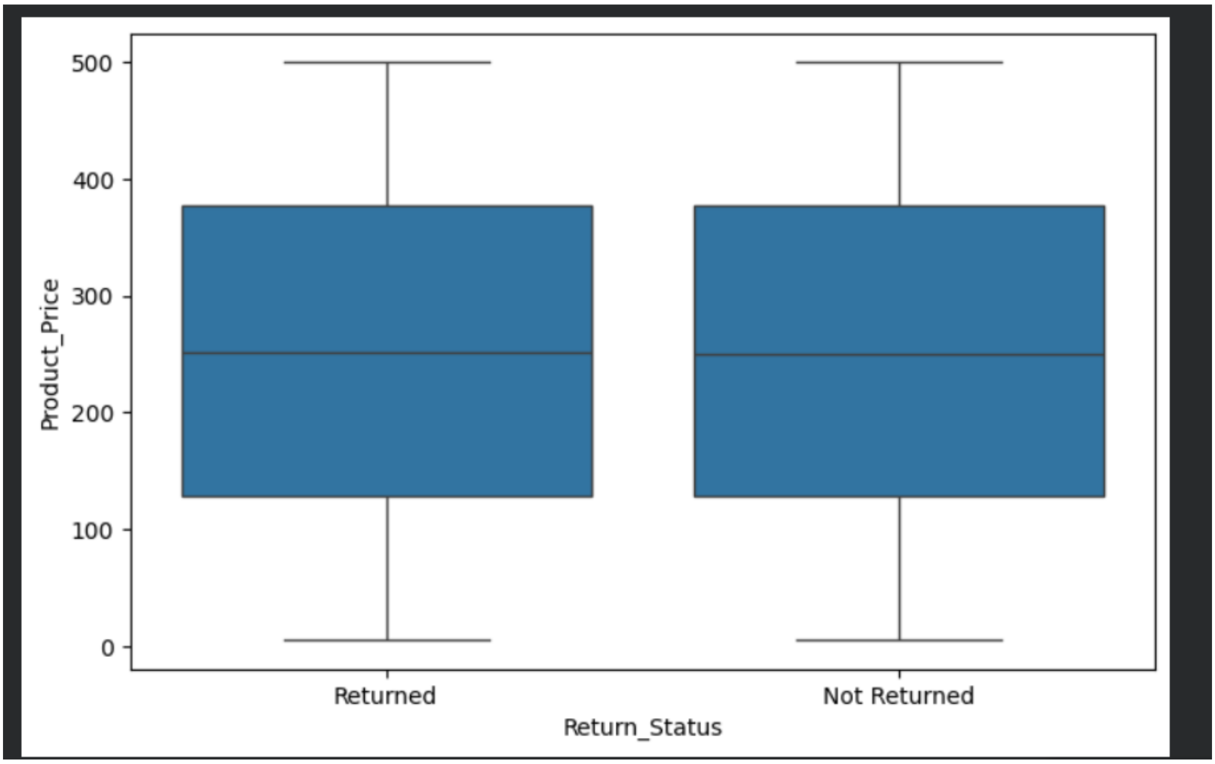
Outlier detection was performed using:

* Boxplots
* Interquartile Range (IQR) Method

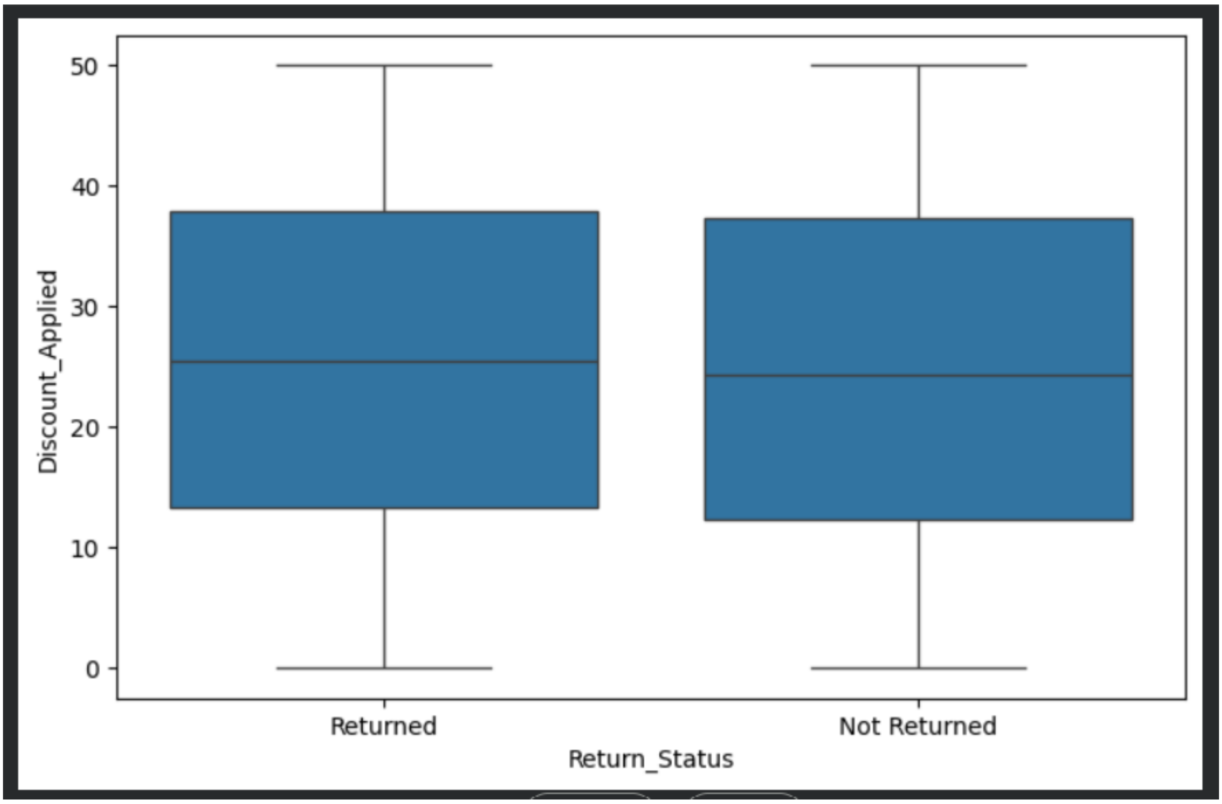
Variables analyzed:

* Product Price
* Discount Applied
* Days to Return

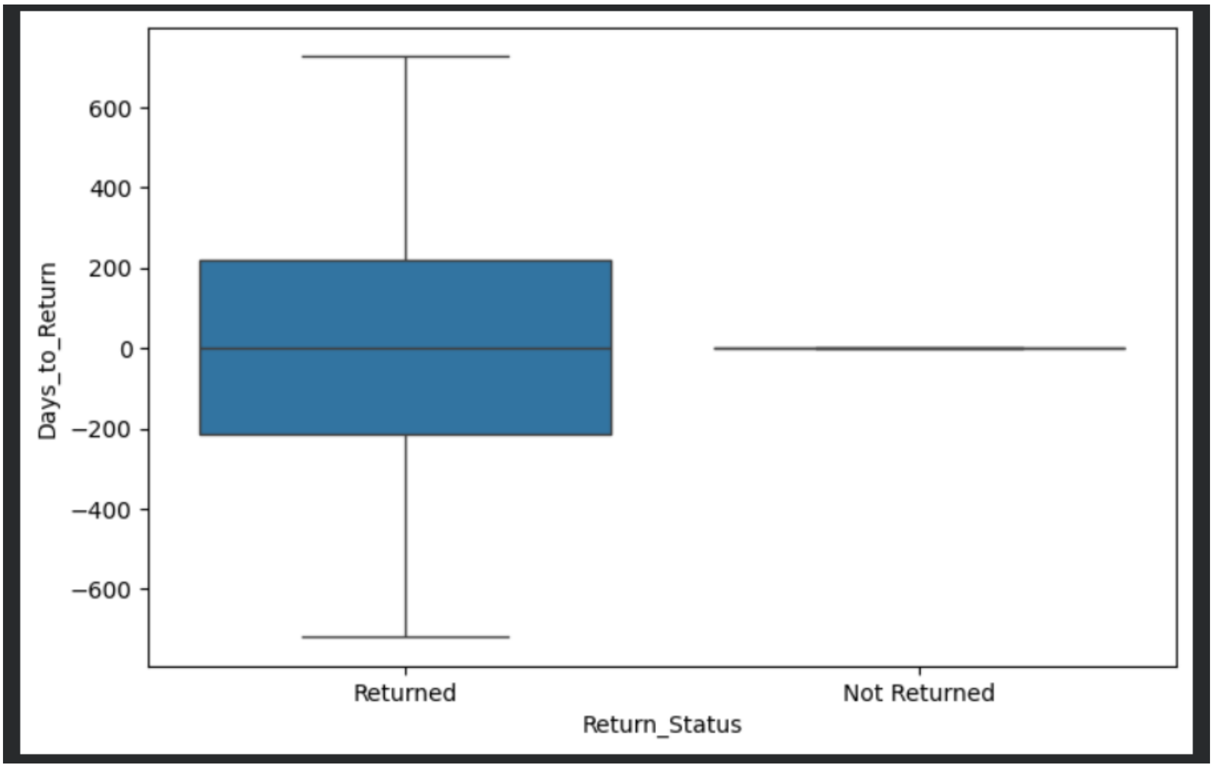
Product_Price vs Return_status:



Discount_applied vs Return_status:



Days_to_return vs Return_status:



HYPOTHESIS TESTING:

Objective:

To determine whether delayed delivery affects product return probability.

Null Hypothesis (H_0):

Delayed delivery has no significant effect on product returns.

Alternative Hypothesis (H_1):

Delayed delivery significantly affects product returns.

Method Used:

Independent Sample T-Test

Result:

- P-value = 0.9369

Conclusion:

Since the p-value is greater than 0.05, we fail to reject the null hypothesis. Therefore, delayed delivery does not significantly affect product return probability in this dataset.

CUSTOMER SEGMENTATION:

Customer segmentation was performed to identify groups that may require business attention based on their purchasing and return behavior. Factors such as product category, discount usage, product price, and return activity were analyzed to understand different customer segments and support targeted business strategies.

BUSINESS RECOMMENDATIONS:

The following recommendations are suggested to reduce product returns and improve customer satisfaction:

1. Improve product quality checks to reduce defective product returns.
2. Strengthen order verification processes to minimize wrong-item deliveries.
3. Provide accurate product descriptions and images to set clear customer expectations.
4. Monitor high-return categories such as Clothing, Books, and Electronics.

5. Review discount strategies and promotional campaigns to understand their impact on returns.
6. Collect and analyze customer feedback regularly to identify recurring issues and improve products and services.

KEY FINDINGS:

1. Product defects are the leading cause of returns.
2. Wrong-item deliveries significantly contribute to return activity.
3. Clothing, Books, and Electronics are the highest-return categories.
4. Delayed delivery does not significantly affect return probability.
5. Customer expectation mismatches remain a major driver of returns.
6. Return behavior is influenced more by product-related issues than by numerical variables such as price or customer age.

CONCLUSION:

The analysis identified product defects, wrong-item deliveries, and customer expectation mismatches as the primary causes of returns. The findings can help e-commerce businesses improve product quality, reduce return rates, and enhance customer satisfaction through data-driven decision-making.